Project So Far 3.0 - BattingEdge FYP

Complete Project History & Current State (Nov 30, 2024)

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1. Project Evolution Timeline

Phase 0-1: Foundation (SUCCESS)

- **Environment Setup**: Python 3.11, VSCode, Git/GitHub, requirements.txt
- **Proof of Concept**: MediaPipe Pose skeleton extraction (33 landmarks)
- **Visual Demo**: skeleton_overlay_demo.mp4 created
- **Status**: 100% Complete

Phase 2-3: The Research Journey (V1-V8p)

V1-V3: Random Forest Experiments (FAILED - 22% accuracy)

- **Approach**: Flattened features (mean, std, max, min of 7 angles)
- **Result**: Massive overfitting (96% train, 22% test)
- **Learning**: Sequential data requires sequential models

V4: Baby LSTM (FAILED - 26% accuracy)

- **Approach**: LSTM on 225 error videos, (50, 7) features
- **Result**: First model to get non-zero F1 scores for minority classes
- **Learning**: LSTM is correct algorithm, but data-starved

V5: Base Model on 1,750 Videos (FAILED - 33% accuracy)

- **Approach**: Bi-LSTM on Kaggle dataset with (50, 7) features
- **Result**: 3.3x better than random but still poor
- **Learning**: Feature starvation - model is "blind" without full skeleton

V6: Transfer Learning Attempt (FAILED - 29% accuracy)

- **Approach**: Fine-tune V5 brain on error detection
- **Result**: Worse than V4 baby model
- **Learning**: V5 learned wrong patterns, making it a "bad teacher"

V7: Full Skeleton Breakthrough (SUCCESS - 70% on 2-shot)

- **Approach**: Complete re-engineering with (50, 66) features
    - All 33 landmarks × 2 coordinates (X, Y)
- **Result**: 70% accuracy on Drive vs Pull diagnostic
- **Learning**: Full skeleton data is critical - 90% improvement over V1-V6

V8p: Production Model (CURRENT - 65% test accuracy)

- **Approach**: Balanced dataset + biomechanical features
- **Architecture**: Bi-LSTM (128→64 units) with dropout
- **Features**: 103 total (99 skeletal + 4 biomechanical)
- **Training**: 78% accuracy on 1,889 balanced videos
- **Test Performance**: 65.3% on real-world data

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2. V8p Model Architecture

Feature Engineering (103 Features)

Skeletal Features (99)

- 33 MediaPipe landmarks × 3 coordinates (x, y, z) = 99 features
- Landmarks include: nose, shoulders, elbows, wrists, hips, knees, ankles, etc.

Biomechanical Features (4)

1. **Wrist Velocity** - Motion intensity indicator
2. **Knee Angle** - Lower body stability measure
3. **Bat Angle Proxy** - Arm orientation relative to shoulder line
4. **Stance Width** - Horizontal distance between feet

Model Structure

Input: (50 timesteps, 103 features)

↓

Bi-LSTM (128 units, return_sequences=True)

↓

Dropout (0.4)

↓

Bi-LSTM (64 units)

↓

Dropout (0.4)

↓

Dense (64, ReLU activation)

↓

Batch Normalization

↓

Dropout (0.3)

↓

Output Dense (4, Softmax) → [Drive, Pull, Cut, Sweep]

Training Configuration

- **Optimizer**: Adam (learning rate: 0.0001)
- **Loss**: Categorical Crossentropy
- **Class Weights**: Applied to handle residual imbalance
- **Early Stopping**: Patience 15, restore best weights
- **Dataset Split**: 80% train, 10% validation, 10% test

Performance Metrics (Test Set - 219 videos)

Shot Class	Precision	Recall	F1-Score	Support	Accuracy
Pull	0.85	0.80	0.82	60	**81.7%**
Cut	0.65	0.62	0.63	61	**62.3%**
Drive	0.58	0.57	0.58	73	**57.5%**

Shot Class	Precision	Recall	F1-Score	Support	Accuracy
Sweep	0.52	0.56	0.54	25	**56.0%**
Overall	0.65	0.64	0.64	219	**65.3%**

Key Insights:

- Pull shots: Highest accuracy (81.7%) due to distinct biomechanics
- Drive-Cut confusion: Main error source (similar front-foot positioning)
- Sweep: Limited by small sample size (only 25 test videos)
- **2.6x better than random baseline (25%)**

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3. Dataset Evolution

Dataset V7 (Initial - IMBALANCED)

- **Total**: 1,517 videos
- **Problem**: Drive bias (381 Drive vs 140 Sweep)
- **Result**: Model collapsed to Drive predictions

Dataset V8p (Current - BALANCED)

- **Total**: 1,889 videos
- **Distribution**: ~380 videos per class (Drive, Pull, Cut, Sweep)
- **Augmentation Applied**:
    - Horizontal flips (simulate left/right-handed batters)
    - Brightness adjustments (±20%)
    - Speed variations (0.9x - 1.1x)
    - Temporal jitter (±2 frames shift)
- **Structure**:
    - train/: 1,520 videos (380 per class)
    - val/: 150 videos (validation split)

- test/: 219 videos (real-world distribution, untouched)

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4. Inference Pipeline Architecture

Dual-Stream Innovation

Problem Solved: Coordinate mismatch between training (full-frame) and inference (cropped/tracked)

Logic Stream (For Model Prediction)

1. Extract MediaPipe landmarks from **full frame**
2. Calculate 103 features using original coordinates
3. Feed to Bi-LSTM model
4. Get shot classification + confidence

Visual Stream (For Overlay Display)

1. Detect person bounding box (MediaPipe Pose detection)
2. Track and crop to person region
3. Extract landmarks relative to **cropped frame**
4. Draw skeleton overlay with HUD
5. Generate output video

Why Separate Streams?

- Training data uses full-frame coordinates
- Inference videos have varying person sizes/positions
- Cropping improves visual presentation but changes coordinate space
- Dual-stream maintains accuracy while enhancing UX

Inference Workflow

python

```
def predict(video_path):
    # Logic Stream
```

```
full_frame_landmarks = extract_pose(video, use_full_frame=True)
features = engineer_features(full_frame_landmarks)  # 103 features
features_50 = resample_to_50_frames(features)
prediction = model.predict(features_50)

# Visual Stream
cropped_landmarks = extract_pose(video, use_cropping=True)
overlay_video = create_overlay(video, cropped_landmarks, prediction)

# Rule-Based Form Analysis
form_analysis = analyze_form(full_frame_landmarks, prediction['shot'])

return {
    'shot': prediction,
    'form_analysis': form_analysis,
    'overlay_video': overlay_video
}
```

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5. Rule-Based Form Analysis System

Overview

Purpose: Provide biomechanical coaching feedback independent of ML model

Architecture: Deterministic rules based on sports science research

Input: 50-frame landmark sequence + predicted shot type

Output: 5 biomechanical checks with pass/fail + recommendations

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Check 1: Front Elbow Angle

Logic

python

 # Calculate angle at impact frame (frame 25 of 50)

shoulder = landmarks[12] *# Right shoulder*

elbow = landmarks[14] *# Right elbow*

wrist = landmarks[16] *# Right wrist*

 # Vectors

v1 = shoulder - elbow

v2 = wrist - elbow

 # Angle calculation

angle = arccos(dot(v1, v2) / (norm(v1) * norm(v2))) * 180/π

Optimal Range: 120° - 140°

- **< 120°**: Elbow too bent → "chicken wing" → reduced power
- **> 140°**: Elbow too straight → loss of control
- **Optimal (120-140°)**: Maximum power transfer + control

Accuracy: ~85%

- Reliable for drives and pulls
- Less reliable for cuts (different elbow mechanics)

Recommendations:

- **Too bent**: "Extend your front arm earlier. Practice shadow batting with focus on full extension at contact."
- **Too straight**: "Allow slight elbow flex for better bat control. Try the '90-120 drill' - start at 90° and extend to 120°."
- **Optimal**: "Excellent elbow positioning! Maintain this form."

Check 2: Head Stability

Logic

python

 # Track nose landmark (landmark 0) across all 50 frames

nose_y_positions = [landmarks[frame][0].y for frame in range(50)]

 # Calculate vertical movement in pixels

max_movement = max(nose_y_positions) - min(nose_y_positions)

 # Convert to centimeters (assuming avg person height = 170cm)

 # Typical head in frame = ~80 pixels = ~25cm

cm_movement = max_movement * (25 / 80)

Optimal Threshold: < 10cm movement

- **< 5cm**: Excellent stability
- **5-10cm**: Acceptable
- **> 10cm**: Head falling/dipping → loss of sight on ball

Accuracy: ~90%

- Very reliable metric
- Clear correlation with shot quality in coaching literature
- Validated against expert coach assessments

Shot-Specific Nuances:

- **Drive**: Strictest threshold (<8cm) - requires still head
- **Pull**: Slightly relaxed (10cm) - some head movement natural
- **Cut**: Medium (9cm)
- **Sweep**: Most relaxed (12cm) - intentional lowering

Recommendations:

- **Excessive movement**: "Your head is dipping/moving too much. Practice the 'ball on head' drill - balance a tennis ball on your cap while shadow batting."
- **Good**: "Head stability looks good. Keep your eyes level throughout the shot."

Check 3: Back Foot Stability

Logic

python

 # Track right ankle (landmark 28) for right-handed batter

initial_ankle_y = landmarks[0][28].y *# First frame position*

 # Check each frame for lifting

for frame in range(50):

  current_ankle_y = landmarks[frame][28].y

  lift_distance = initial_ankle_y - current_ankle_y *# Negative = lifting*

  if lift_distance < -5cm: *# Back foot lifted more than 5cm*

    is_error = True

Optimal Behavior: Back foot stays grounded

- **< 2cm lift**: Excellent anchor
- **2-5cm lift**: Acceptable for power shots
- **> 5cm lift**: Base instability → poor weight transfer

Accuracy: ~75%

- Can have false positives if camera angle changes
- Reliable in controlled environments
- Issues with low frame-rate videos (<24fps)

Shot-Specific Rules:

- **Drive**: Must stay grounded (strict)
- **Pull**: Allowed to lift slightly (relaxed to 7cm) - natural for power
- **Cut**: Should stay grounded
- **Sweep**: Can lift (intentional lowering + rotation)

Recommendations:

- **Early lifting**: "Your back foot is lifting too early, causing balance issues. Practice the 'anchor drill' - place a weight on your back foot during shadow practice."
- **Good**: "Back foot stability is solid. Good foundation for power transfer."

Check 4: Hip Rotation

Logic

python

 # Calculate hip line angle (landmarks 23=left hip, 24=right hip)

 # Compare initial frame vs impact frame (frame 25)

```
def calculate_hip_angle(left_hip, right_hip):
    # Vector from right hip to left hip
    hip_vector = left_hip - right_hip
    # Angle relative to horizontal
    angle = arctan2(hip_vector.y, hip_vector.x) * 180/π
    return angle

initial_angle = calculate_hip_angle(landmarks[0][23], landmarks[0][24])
impact_angle = calculate_hip_angle(landmarks[25][23], landmarks[25][24])

rotation = abs(impact_angle - initial_angle)
```

Shot-Specific Optimal Ranges:

- **Drive**: 30°+ (moderate rotation for timing)
- **Pull**: 60°+ (aggressive rotation for power)
- **Cut**: 20°+ (minimal rotation, more arms)
- **Sweep**: 45°+ (full body rotation)

Accuracy: ~70%

- Challenging due to camera angle dependency
- 2D projection of 3D rotation loses information
- Works best with side-on camera angles

Recommendations by Shot:

- **Drive (insufficient)**: "Rotate your hips more through the shot. Try the 'pivot drill' - practice hip turn without the bat."
- **Pull (insufficient)**: "Pull shots need aggressive hip rotation. You're rotating only [X]° but need 60°+. Practice pull shots against spin bowling to build rotation habit."
- **Optimal**: "Excellent hip rotation! You're generating good power through the lower body."

Check 5: Follow-Through Height

Logic

python

 # Check final frame (frame 49) position

right_wrist_y = landmarks[49][16].y

right_shoulder_y = landmarks[49][12].y

 # Wrist should finish higher than shoulder

height_difference = right_shoulder_y - right_wrist_y *# Positive = good*

```
if height_difference > 0:
    is_error = False  # Hands finished high
else:
    is_error = True   # Incomplete follow-through
```

Optimal Behavior: Hands finish above shoulder height

- **> 20cm above shoulder**: Excellent, full follow-through
- **5-20cm above**: Acceptable
- **< 5cm or below**: Incomplete follow-through → reduced power

Accuracy: ~80%

- Very reliable for drives and lofted shots
- Less applicable to cuts (different mechanics)
- Can have issues with camera framing (if follow-through goes off-screen)

Shot-Specific Adjustments:

- **Drive**: Strict (must finish high)
- **Pull**: Very strict (power shot needs full follow-through)
- **Cut**: Relaxed (hands may finish lower)
- **Sweep**: Medium (follow-through arc different)

Recommendations:

- **Incomplete**: "Your follow-through is incomplete. You're stopping at [X]cm below shoulder. Practice 'full swing' drills - exaggerate the follow-through to build muscle memory."
- **Good**: "Complete follow-through! You're maximizing power transfer."

Form Score Calculation

python

```
def calculate_form_score(checks):
    score = 100
```

```
    for check in checks:
        if check['severity'] == 'minor':
            score -= 10
        elif check['severity'] == 'major':
            score -= 20

    return max(0, score)  # Floor at 0
```

Severity Classification:
- **Minor Error**: Metric slightly outside optimal range (5-15% deviation)
- **Major Error**: Metric significantly outside optimal range (>15% deviation)

Example:
- Perfect form: 100/100
- One minor error (head movement 11cm): 90/100
- One major error (elbow angle 105°): 80/100
- Two major errors: 60/100

Limitations & Future Improvements

*#### **Current Limitations**:*

1. **2D Analysis**: Loses depth information (hip rotation most affected)

2. **Camera Angle Dependency**: Works best with side-on angles

3. **Frame Rate Sensitivity**: Requires minimum 24fps for accuracy

4. **No Bat Tracking**: Cannot analyze bat path, speed, angle

*#### **Planned Enhancements (Finals)**:*

1. **YOLOv8 Bat Detection**: Add bat angle, speed, trajectory features

2. **3D Pose Estimation**: Use stereo cameras or depth sensors

3. **Dynamic Thresholds**: Learn optimal ranges per user over time

4. **Swing Path Analysis**: Track bat trajectory arc

5. **Impact Point Detection**: Identify exact contact frame

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6. Key Milestones Completed

✅ **Dataset**: 1,889 balanced videos across 4 shot types ✅ **Features**: 103-dimensional feature engineering (99 skeletal + 4 biomechanical) ✅ **Model**: Bi-LSTM achieving 65% test accuracy (78% training) ✅ **Inference**: Dual-stream architecture with form analysis ✅ **Validation**: 75% accuracy on validation set (matches training distribution) ✅ **Demo Videos**: 6 production-ready overlay videos for defense ✅ **Rule-Based System**: 5 biomechanical checks with coaching recommendations